Supplementary Material: Uncertainty-Aware Open-Vocabulary Gaussian Splatting for Changing Scenes

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This supplement accompanies the paper. Numbers of sections, tables and figures without the prefix S refer to the paper. Every number comes from a saved evaluation of the paper's protocol. The test scenes had been evaluated before the analyses of Sections S1, S2 and S5, so their tests are descriptive; Section S6 lists the later attempts to improve the results, each registered before it ran.

S1. Calibration and paired tests

All test scenes were seen before these tests, so they are descriptive. For each baseline and metric of Table I we count the test scenes in which ours is better, take the median paired difference, and apply an exact two-sided sign test over the 14 scenes of both datasets, with Holm's correction over the 20 comparisons (Table S1). Ours and averaging do not differ in any metric; ours is better than LEGaussians in every metric and has a lower NLL than Feature 3DGS and lower ECE, NLL and Brier than the 2D features in 13 of 14 scenes. Averaging and P0 have the same lead over the trained rows, so it comes from training-free lifting, not from the adaptive temperature. ECE also depends on how it is averaged (Table S2, Fig. S1): over all test pixels pooled, P2 has the highest ECE of the open-vocabulary rows on Replica (7.1), while it has the lowest mean over scenes (8.7); its Brier score is the lowest under both.

S2. Stale content

A. Fused baselines

Table S3 repeats the fused rows of Table II and tests our fused row against each by an exact sign test over the ten test scenes, with Holm's correction over the eight tests. The feature residual and NormLift's drop, fused with depth the same way as ours, rank the stale Gaussians better than our fused row.

B. Measured false-alarm rate

The threshold targets a per-view false-alarm rate of 10^{-3} on static content (Sec. III-F). Table S4 gives the measured rate r_{static}, the share of the tests on static observed Gaussians during the update that fire, for every row at its own label-free threshold. Ours has a median of 4.0×10^{-3} over the ten test scenes, within a factor of two of nominal in one scene; it is closer to nominal than the feature residual, NCC, the dispersion and NormLift's drop in 10, 9, 9 and 8 of the ten scenes, and less close than the depth residual (3 of 10).

C. Qualitative results and stale answers

Fig. S2 renders every change score on three test scenes chosen by a fixed rule, and Fig. S3 shows how the semantics of a text query change over the update. The stale-answer rate of Sec. IV-D is the share of the removed object's covered pixels, over all update keyframes, whose top class over the 101 Replica classes is the scene's query (the most frequent top class of the stale Gaussians after mapping, computed by rule).

S3. Static scenes: per class and per method

Fig. S4 shows both uncertainties on a Replica test frame. Table S5 gives the per-class IoU of the rows of Table I. On Replica, ours leads on ceiling and chair, and the other rows lead elsewhere by 0.1 to 2.1 points. On ScanNet, ours leads on cabinet, chair and bed and ties on window, while averaging leads on floor, wall and door by 0.2 to 0.4 points. Fig. S5 compares every method on one test frame of each dataset, and Fig. S6 on one frame of every test scene; scenes and frames follow rules fixed before rendering.

S4. Agreement estimator

Fig. S13 illustrates the estimator (2) of Sec. III-E on simulated views: it keeps little evidence and conflicting evidence apart, which the mean resultant length and the prior-shrunk agreement do not.

S5. Fitted temperatures and build cost

Table S6 lists the fitted P2 coefficients; on ScanNet γ_e and γ_a are negative. Table S7 gives the build time per ScanNet scene from the production logs of a shared RTX 4070 Ti, so the times include contention with other jobs.

Run time per keyframe. On one Replica validation scene (NVIDIA RTX 4070 Ti, Intel Core i7-13700KF, no other job), a keyframe takes 109.9 ± 7.6 ms: 69.9 ± 2.7 ms for LSeg, 20.4 ± 6.0 ms for the render and backward pass, and 16.7 ± 2.4 ms for the statistics, change test and re-admission. Peak GPU memory is 1.9 GiB; averaging and NormLift reuse our sums and add 2.3 ms; the field state takes 844 bytes per Gaussian. Fitting a Replica map took 4.5 ± 0.6 min on an RTX 5070 Ti.

TABLE S1

OURS AGAINST EACH BASELINE, PAIRED OVER THE TEST SCENES (REPLICA $N = 6$, SCANNET $N = 8$). X/N: SCENES IN WHICH OURS IS BETTER (MEDIAN PAIRED DIFFERENCE IN BRACKETS; POSITIVE: OURS BETTER; POINTS EXCEPT NLL IN NATS). MIOU COMPARES OURS P0, THE OTHER COLUMNS OURS P2. SIGN TEST OVER THE 14 SCENES, HOLM OVER THE 20 TESTS (**BOLD**: BELOW 0.05); WILCOXON AS A SECONDARY COLUMN.

Baseline	Metric	Replica	ScanNet	Both	Median	Sign p	Holm p / Wilcoxon p
Averaging	mIoU	3/6 (-0.01)	5/8 (0.00)	8/14	0.00	0.79	1 / 0.86
Averaging	ECE	3/6 (+0.32)	3/8 (-0.18)	6/14	-0.18	0.79	1 / 0.71
Averaging	NLL	5/6 (+0.009)	4/8 (-0.002)	9/14	+0.006	0.42	1 / 0.36
Averaging	Brier	6/6 (+0.38)	4/8 (+0.02)	10/14	+0.24	0.18	1 / 0.17
Averaging	AURC	6/6 (+1.24)	0/8 (-0.49)	6/14	-0.09	0.79	1 / 0.86
Feature 3DGS	mIoU	2/6 (-0.17)	7/8 (+1.85)	9/14	+0.49	0.42	1 / 0.078
Feature 3DGS	ECE	4/6 (+0.42)	7/8 (+1.24)	11/14	+0.81	0.057	0.46 / 0.0085
Feature 3DGS	NLL	6/6 (+0.051)	7/8 (+0.033)	13/14	+0.038	0.0018	**0.033** / 0.0031
Feature 3DGS	Brier	6/6 (+0.80)	6/8 (+0.76)	12/14	+0.80	0.013	0.14 / 0.0052
Feature 3DGS	AURC	6/6 (+1.80)	6/8 (+0.16)	12/14	+0.69	0.013	0.14 / 0.0052
LEGaussians	mIoU	6/6 (+3.16)	8/8 (+5.01)	14/14	+3.52	0.00012	**0.0024** / 0.00012
LEGaussians	ECE	6/6 (+1.14)	8/8 (+1.58)	14/14	+1.26	0.00012	**0.0024** / 0.00012
LEGaussians	NLL	6/6 (+0.136)	7/8 (+0.081)	13/14	+0.108	0.0018	**0.033** / 0.00024
LEGaussians	Brier	6/6 (+3.33)	7/8 (+2.14)	13/14	+3.24	0.0018	**0.033** / 0.00024
LEGaussians	AURC	6/6 (+3.37)	7/8 (+0.38)	13/14	+1.62	0.0018	**0.033** / 0.0012
2D features	mIoU	4/6 (+1.05)	7/8 (+5.43)	11/14	+3.45	0.057	0.46 / 0.0085
2D features	ECE	5/6 (+3.50)	8/8 (+2.67)	13/14	+3.21	0.0018	**0.033** / 0.00061
2D features	NLL	6/6 (+0.105)	7/8 (+0.094)	13/14	+0.100	0.0018	**0.033** / 0.00085
2D features	Brier	6/6 (+3.77)	7/8 (+3.55)	13/14	+3.56	0.0018	**0.033** / 0.00037
2D features	AURC	6/6 (+3.49)	6/8 (+1.24)	12/14	+2.01	0.013	0.14 / 0.00061

TABLE S2

CALIBRATION UNDER TWO AGGREGATIONS: MEAN OVER THE TEST SCENES (AS TABLE I) AND POOLED OVER ALL TEST PIXELS. ECE IN %, NLL IN NATS.

	Replica				ScanNet			
	ECE		NLL		ECE		NLL	
Row	mean	pool	mean	pool	mean	pool	mean	pool
Ours P2	8.70	7.12	0.996	0.982	6.06	2.62	0.509	0.506
Ours P0	9.86	5.62	0.996	0.981	6.36	3.16	0.510	0.502
Ours P1	10.21	8.69	1.039	1.022	6.97	3.59	0.522	0.508
Averaging	9.62	5.70	1.004	0.989	6.17	3.13	0.512	0.499
Feature 3DGS	10.01	6.29	1.044	1.028	7.38	4.72	0.571	0.553
LEGaussians	10.59	6.20	1.136	1.121	8.10	4.63	0.628	0.605
2D features	11.78	7.05	1.116	1.097	8.78	5.11	0.638	0.621
CSS (closed set)	5.05	3.13	1.014	0.999	7.03	2.63	0.604	0.556

S6. REGISTERED LATER ATTEMPTS

After the first version of the results, we tried five changes meant to improve them. Each was registered before it ran, chosen on the validation scenes by a rule fixed in advance and, where validation selected it, run once on the test scenes. None improved on the paper's method, which therefore stays unchanged; Table S8 gives the outcomes.

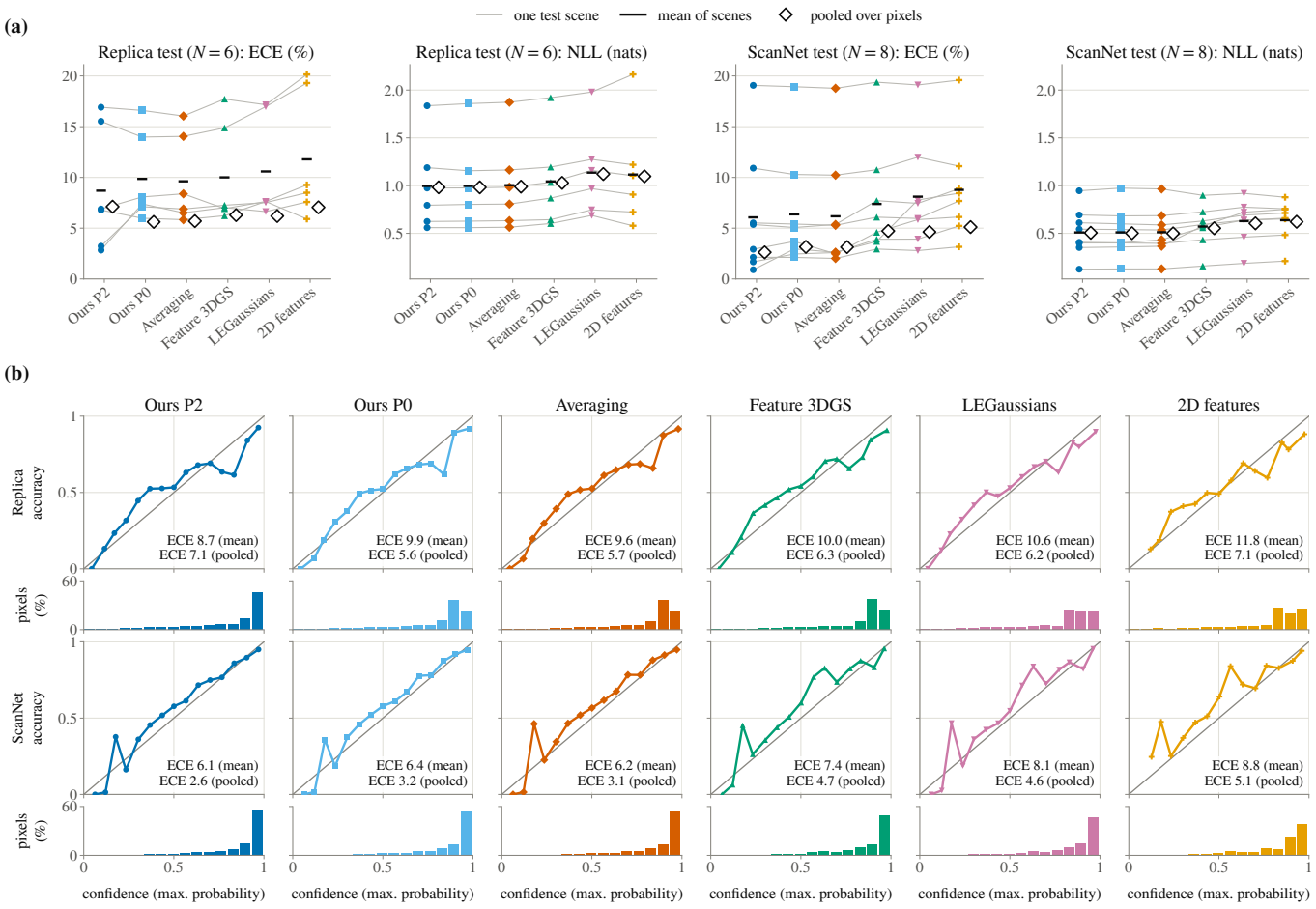


Fig. S1. Calibration on the test scenes. (a) ECE and NLL per scene (grey lines join the rows of one scene), their mean over scenes (bar) and the value pooled over all pixels (diamond). (b) Reliability diagrams pooled over the test pixels of each dataset (15 equal-width bins) with the share of pixels per bin.

TABLE S3

EVERY CHANGE ROW FUSED WITH THE DEPTH RESIDUAL BY THE RANK AVERAGE OF OUR FUSED ROW (PRE-REGISTERED, RUN AFTER TABLE II), OBJECT-REMOVAL TEST SCENES, MEAN ± STD ($N = 10$), %. BOTTOM: OUR FUSED ROW AGAINST EACH, SCENES IN WHICH OURS IS HIGHER, MEDIAN DIFFERENCE (POINTS), SIGN TEST AND HOLM OVER THE 8 TESTS.

		Gaussian level		Pixel level	
Method	In	AP	AUROC	AP	AUROC
Ours fused with depth residual	F+D	75.0 ± 12.9	92.3 ± 2.9	61.8 ± 13.8	78.2 ± 11.3
NCC photometric residual + depth	I+D	61.2 ± 17.7	85.3 ± 9.4	56.8 ± 12.0	74.2 ± 12.3
Feature residual, $k = 2$, + depth	F+D	**81.1** ± 9.4	92.8 ± 3.2	64.3 ± 13.4	78.2 ± 12.1
NormLift reliability drop + depth	F+D	79.9 ± 15.1	**93.9** ± 2.4	**66.9** ± 12.2	**79.6** ± 12.1
Dispersion $1 - \hat{\rho}$ + depth	F+D	34.2 ± 8.7	82.5 ± 5.3	35.8 ± 22.2	61.7 ± 16.2
Ours + depth vs	Level	Higher	Median	Sign p	Holm p
NCC photometric residual	Gaussian AP	9/10	+10.9	0.021	0.107
NCC photometric residual	Pixel AP	8/10	+7.4	0.109	0.328
Feature residual, $k = 2$	Gaussian AP	0/10	-1.2	0.002	0.016
Feature residual, $k = 2$	Pixel AP	3/10	-0.6	0.344	0.344
NormLift reliability drop	Gaussian AP	1/10	-1.7	0.021	0.107
NormLift reliability drop	Pixel AP	2/10	-4.8	0.109	0.328
Dispersion $1 - \hat{\rho}$	Gaussian AP	10/10	+41.2	0.002	0.016
Dispersion $1 - \hat{\rho}$	Pixel AP	10/10	+25.5	0.002	0.016

TABLE S4

MEASURED PER-VIEW FALSE-ALARM RATE ON STATIC GAUSSIANS, r_{static} ($\times 10^{-3}$; NOMINAL 1), ON THE OBJECT-REMOVAL SCENES. MEDIAN (MIN–MAX) OVER THE TEN TEST SCENES, POOLED OVER THEM, VALIDATION MEDIAN, MEDIAN $|\log_{10}(r_{\text{static}}/10^{-3})|$, TEST SCENES WITHIN A FACTOR OF TWO OF NOMINAL, AND TEST SCENES IN WHICH OURS IS CLOSER TO NOMINAL THAN THE ROW.

Row	Test median (range)	Test pooled	Val. median	$\lvert\log_{10}\rvert$	Within 2×	Ours closer
NCC photometric residual	22.2 (1.1–204.9)	32.1	32.9	1.35	1/10	9/10
Depth residual	1.1 (0.3–9.1)	2.6	1.6	0.40	3/10	3/10
Feature residual, $k = 2$	52.8 (4.3–277.8)	71.0	68.0	1.72	0/10	10/10
NormLift reliability drop	4.1 (0.1–191.0)	50.7	6.6	0.90	0/10	8/10
Dispersion $1 - \hat{\rho}$ (ours)	8.3 (3.0–68.4)	28.8	2.3	0.91	0/10	9/10
Ours: test, reset, re-admission	4.0 (0.5–23.4)	3.6	4.8	0.60	1/10	–

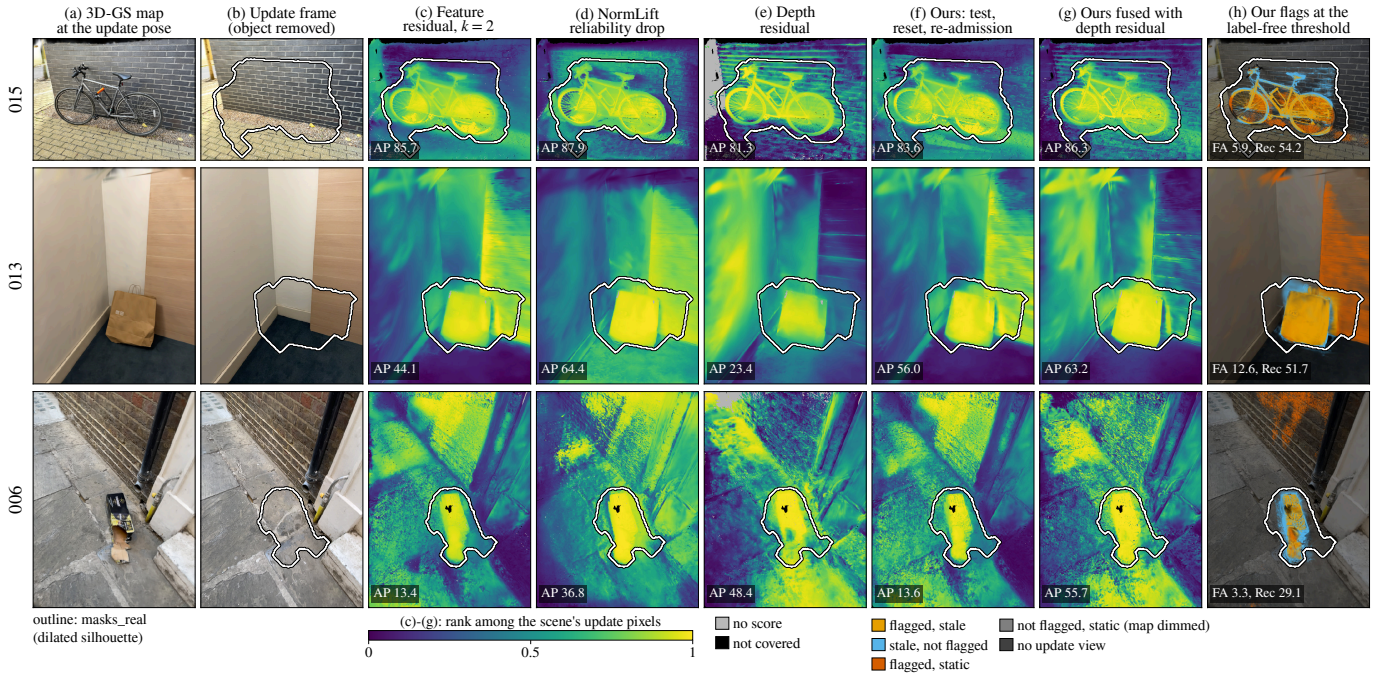


Fig. S2. Stale content after a real removal (NeRF-Object-Removal test scenes 015 at frame 675, 013 at frame 556, 006 at frame 178). The scenes are fixed by rule: the highest, the lower-median and the lowest pixel-level AP of ours in Table II; the frame is the update keyframe with the largest object region inside the image. The map geometry is fixed: (a) renders the map, so the removed object's Gaussians are still there; (b) is the camera image after the removal. (c)-(g) render each row's per-Gaussian score, coloured by its rank among the covered pixels of all update keyframes of the scene (the population of the pixel-level AP; grey: no score, black: not covered); the numbers are the rows' pixel-level AP (%) in Table II. (h) shows our flags at the label-free threshold against the stale Gaussians (orange: flagged and stale, light blue: stale and not flagged, vermillion: flagged and static), with FA/Rec. (%) of ours counted over all update views (015 5.9/54.2, 013 12.6/51.7, 006 3.3/29.1). The outline is masks_real, a dilated silhouette of the object. NormLift's drop has a higher pixel-level AP than our test in all three scenes, as in Table II (9 of 10 test scenes).

TABLE S5

PER-CLASS IOU (%) ON THE REPLICA ($N = 6$) AND SCANNET ($N = 8$) TEST SCENES FOR THE CLASSES IN AT LEAST FIVE SCENES (MEAN OVER THE SCENES WITH THE CLASS; ON REPLICA, VENT AND WALL-PLUG SCORE 0 FOR EVERY ROW AND ARE LEFT OUT). MIOU: OVER ALL CLASSES (TABLE I). MARKS PER DATASET AS IN TABLE I; AN IOU OF 0 IS NOT MARKED; CSS IS NOT RANKED.

	Replica										ScanNet							
Method	ceiling	door	floor	lamp	wall	chair	panel	table	window	**mIoU**	floor	wall	cabinet	door	chair	bed	window	**mIoU**
2D features, per frame	78.9	47.0	46.9	25.2	75.9	63.2	0.0	51.3	12.3	26.3	88.6	79.1	66.1	56.3	58.5	80.4	54.0	62.5
Averaging [1]†	80.5	**60.0**	49.6	22.4	78.0	70.0	0.0	**53.8**	15.6	27.9	**90.3**	**81.6**	71.6	**67.3**	68.9	89.7	**57.0**	67.7
Feature 3DGS [2]†	80.8	59.2	**49.7**	25.0	**78.1**	65.6	0.0	52.4	**17.1**	27.9	89.7	81.0	69.6	62.6	63.9	83.6	55.7	65.0
LEGaussians [3]†	79.0	52.5	45.8	**25.9**	75.7	58.9	**0.1**	48.4	15.6	24.0	88.3	78.7	66.6	56.7	59.9	81.5	53.3	62.3
Ours (P0 and P2 share labels)	**80.9**	59.7	49.3	23.8	78.0	**70.7**	0.0	53.1	15.6	**28.0**	90.1	81.3	**72.0**	66.9	**70.4**	**90.1**	**57.0**	**68.0**
CSS [4]†, closed set	80.5	54.9	50.3	27.0	78.3	68.6	0.0	54.1	16.2	28.5	90.5	82.1	72.4	68.3	66.5	88.8	57.1	67.3

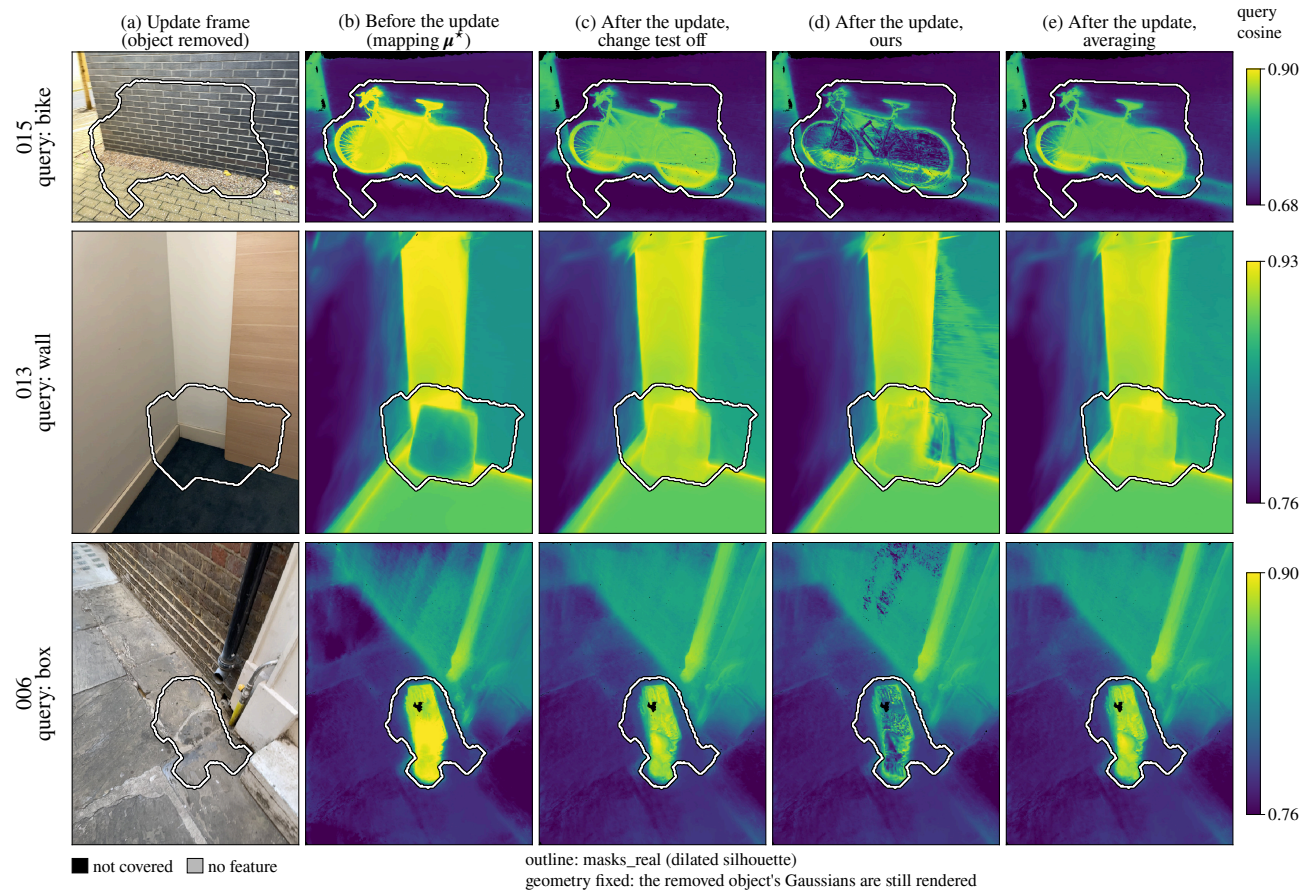


Fig. S3. Query relevance before and after the update, same scenes, frames and rotations as Fig. S2. The query of each scene is fixed by rule and never re-prompted: the most frequent top class (101 Replica classes, LSeg 5-template ensemble) of the mapping-pass $\mu_n^\star$ of the stale Gaussians (015: 'bike', 013: 'wall', 006: 'box'). Each panel shows the cosine between the rendered feature and the query text, one colour scale per row (5th to 99th percentile of (b)). The map geometry is fixed: the removed object's Gaussians are still rendered after the update and only their semantics change. (c) is the same field after the same update views with the change test off; (d) is ours, where only the Gaussians our test flags are reset (Rec. 54.2/51.7/29.1% in these scenes), so the missed ones keep the old class; (e) is averaging. The outline is masks_real, a dilated silhouette.

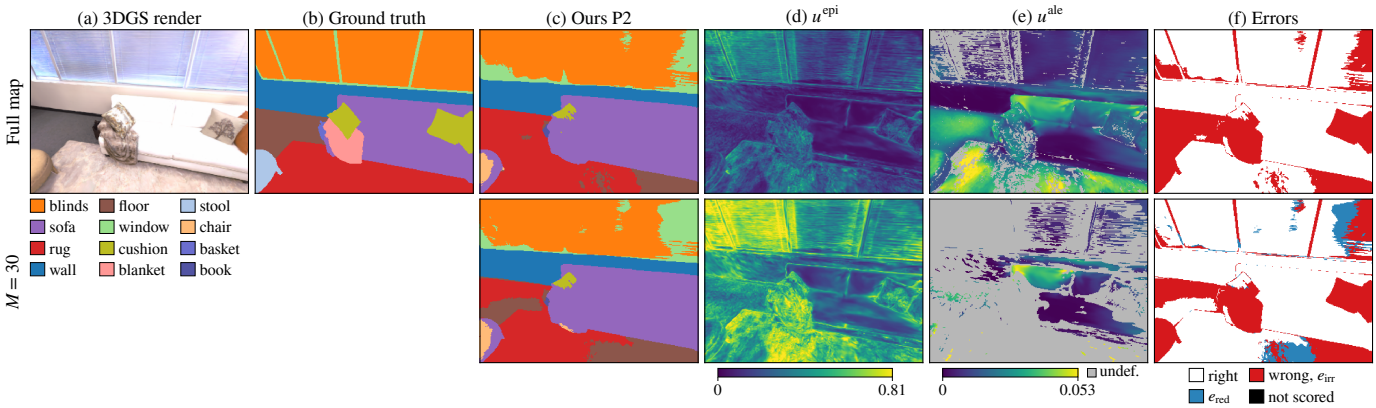


Fig. S4. Uncertainty on a Replica test frame (room_0, the evaluation frame with the most reducible-error pixels). Top: the full map; bottom: a field from $M = 30$ of the 180 mapping frames on the same geometry. (c) labels of P2, (d) u^{epi}, (e) u^{ale} (grey: undefined, fewer than $\tau_{\min}$ views), (f) right and wrong pixels; at $M = 30$ the wrong pixels split into reducible (e_{red}, blue) and irreducible (e_{irr}, red) errors. With 30 frames, u^{epi} rises, also where the reducible errors are, while u^{ale} is still undefined on most pixels; with the full map, u^{ale} is highest at the rug, where the labels stay wrong.

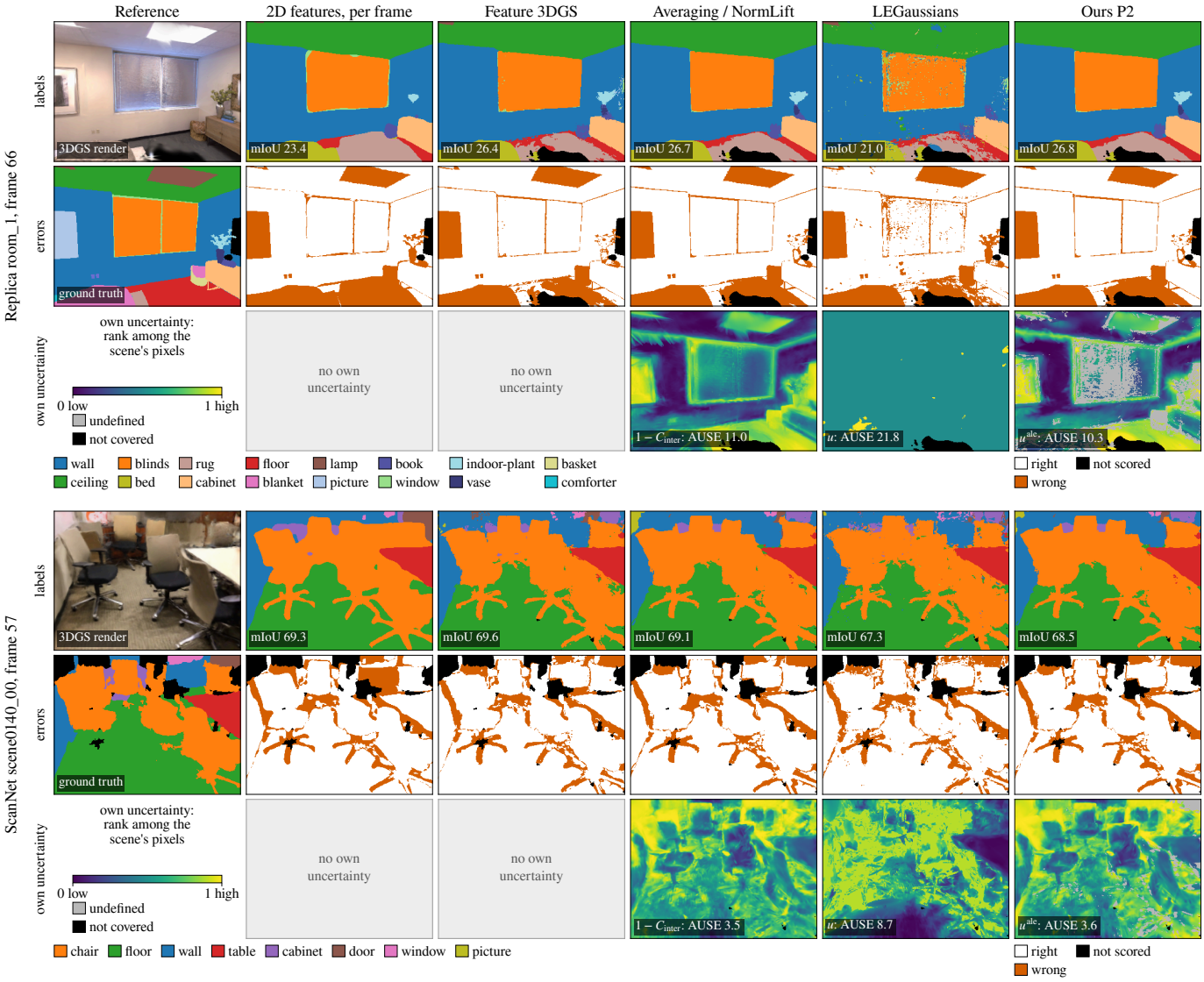


Fig. S5. Segmentation and own uncertainty of every method on Replica room_1 (frame 66) and ScanNet scene0140_00 (frame 57). Each scene is the test scene with the lower-median per-scene mIoU of ours P2 in Table I, and each frame is the evaluation frame with the most ground-truth classes that cover at least 0.5% of its labeled pixels; both rules read only the published table and the labels and were fixed before rendering. Row 1: the 3DGS render and the labels (black: no decision); row 2: the ground truth and the errors (vermillion: wrong; black: not scored); row 3: each method's own uncertainty, NormLift's $1 - C_{\text{inter}}$, the learned u of LEGaussians and our u^{ale}, as its rank among the covered labeled pixels with a defined value in all evaluation frames of the scene (grey: undefined; black: not covered). The 2D features and Feature 3DGS have no own uncertainty (– in Table I); we show no P2 confidence. Numbers are per-scene values over all evaluation frames, read from Table I: mIoU (row 1) and the AUSE of the own uncertainty (row 3), in %. LEGaussians $\lambda_u = 4$ (Replica), 1 (ScanNet), chosen on validation. On room_1, ours has the highest mIoU (26.8). On room_1, u^{ale} has the lowest AUSE of the own uncertainties (10.3 vs 11.0, 21.8). On scene0140_00, ours (68.5 mIoU) is below Feature 3DGS (69.6), the 2D features (69.3) and averaging (69.1). On scene0140_00, NormLift's $1 - C_{\text{inter}}$ ranks the errors slightly better than u^{ale} (AUSE 3.5 vs 3.6).

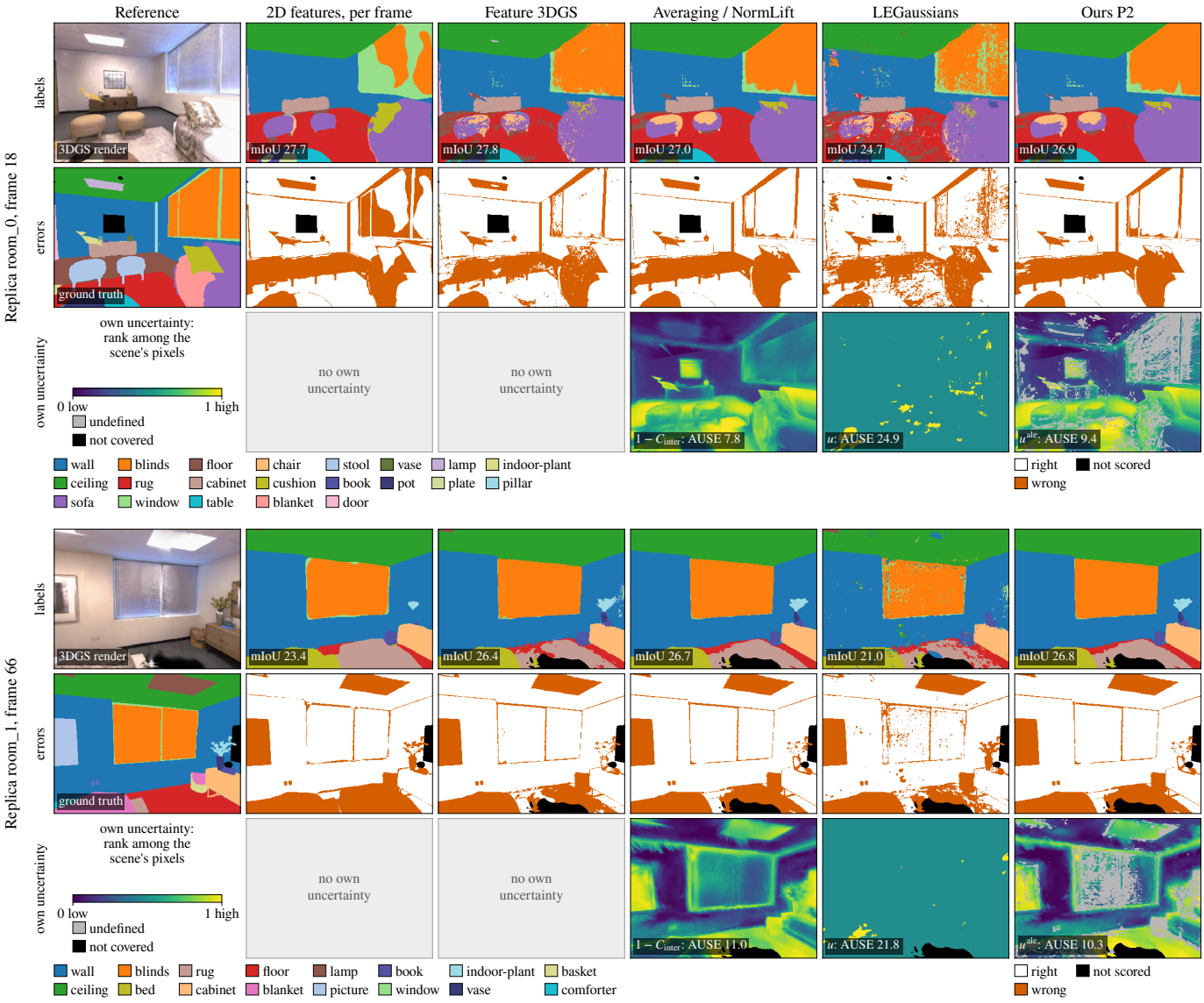


Fig. S6. Segmentation and own uncertainty of every method on Replica room_0 (frame 18), Replica room_1 (frame 66), Replica office_1 (frame 25), Replica office_2 (frame 75), Replica office_3 (frame 15), Replica office_4 (frame 29), ScanNet scene0070_00 (frame 46), ScanNet scene0097_00 (frame 2), ScanNet scene0140_00 (frame 57), ScanNet scene0200_00 (frame 36), ScanNet scene0347_00 (frame 0), ScanNet scene0400_00 (frame 0), ScanNet scene0590_00 (frame 104) and ScanNet scene0645_00 (frame 156). Every test scene, at the evaluation frame with the most ground-truth classes that cover at least 0.5% of its labeled pixels (a rule on the labels only, fixed before rendering). Row 1: the 3DGS render and the labels (black: no decision); row 2: the ground truth and the errors (vermillion: wrong; black: not scored); row 3: each method's own uncertainty, NormLift's $1 - C_{\text{inter}}$, the learned u of LEGaussians and our u^{ale}, as its rank among the covered labeled pixels with a defined value in all evaluation frames of the scene (grey: undefined; black: not covered). The 2D features and Feature 3DGS have no own uncertainty (– in Table I); we show no P2 confidence. Numbers are per-scene values over all evaluation frames, read from Table I: mIoU (row 1) and the AUSE of the own uncertainty (row 3), in %. LEGaussians λ_u = 4 (Replica), 1 (ScanNet), chosen on validation.

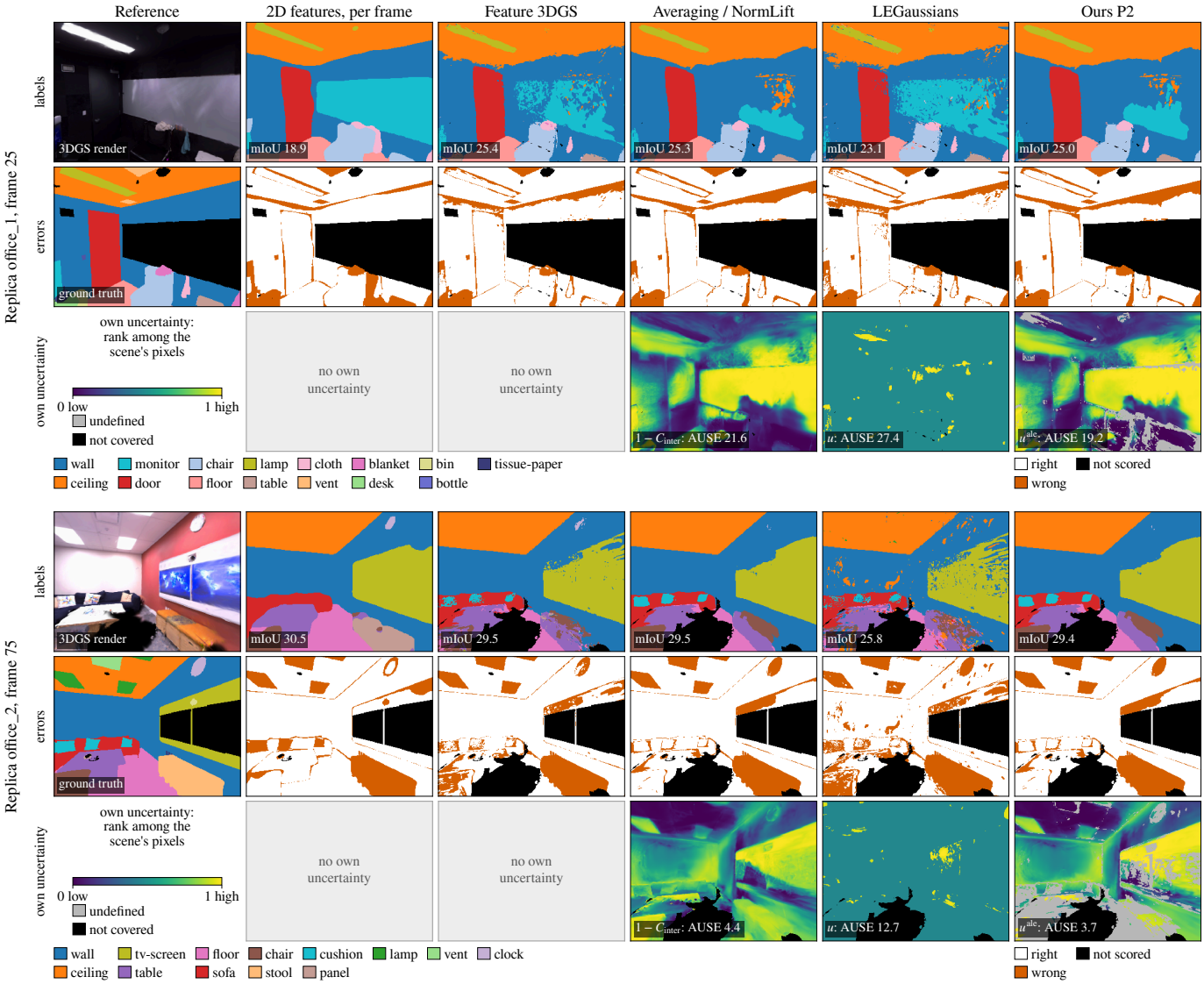


Fig. S7. Continued from Fig. S6.

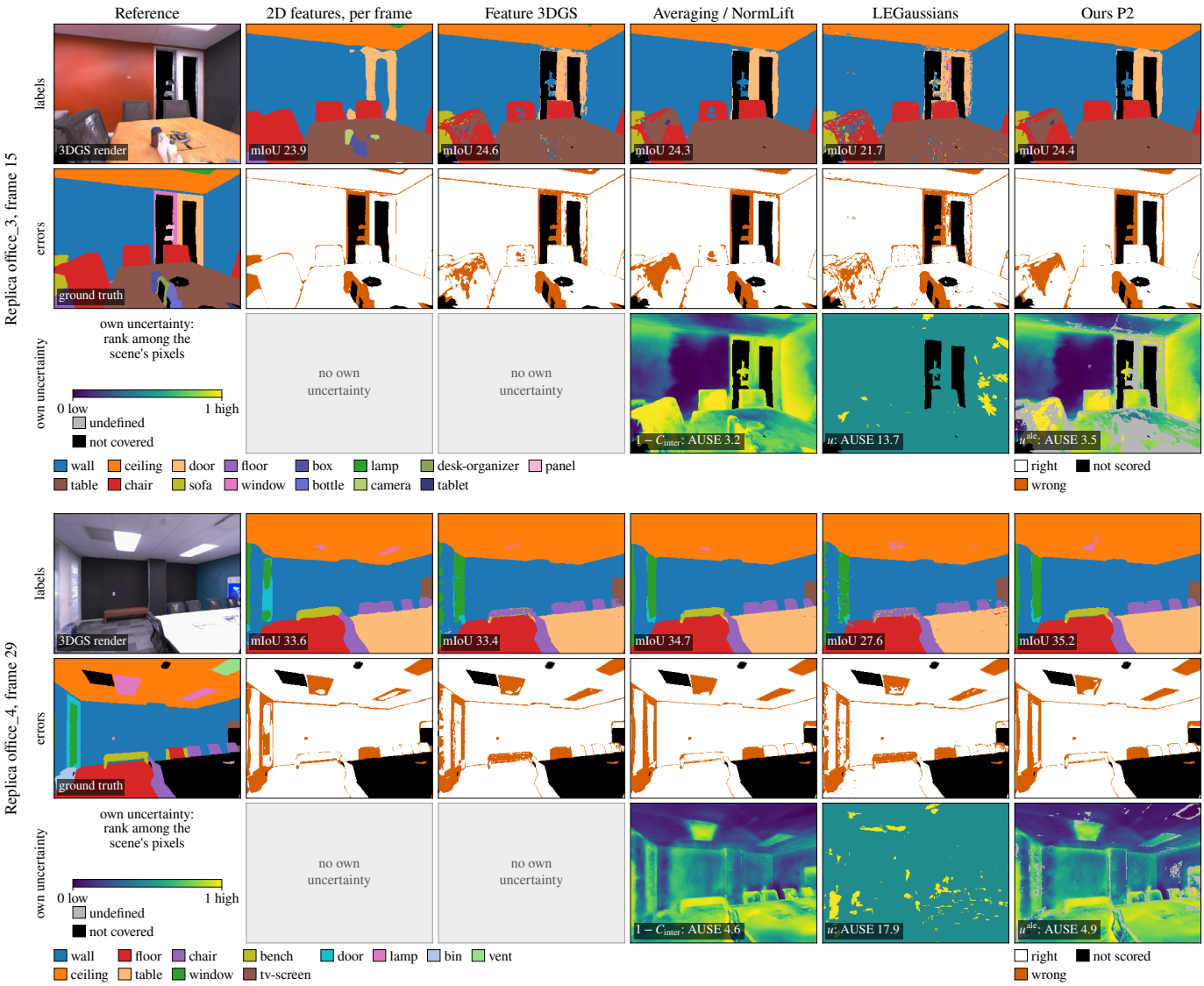


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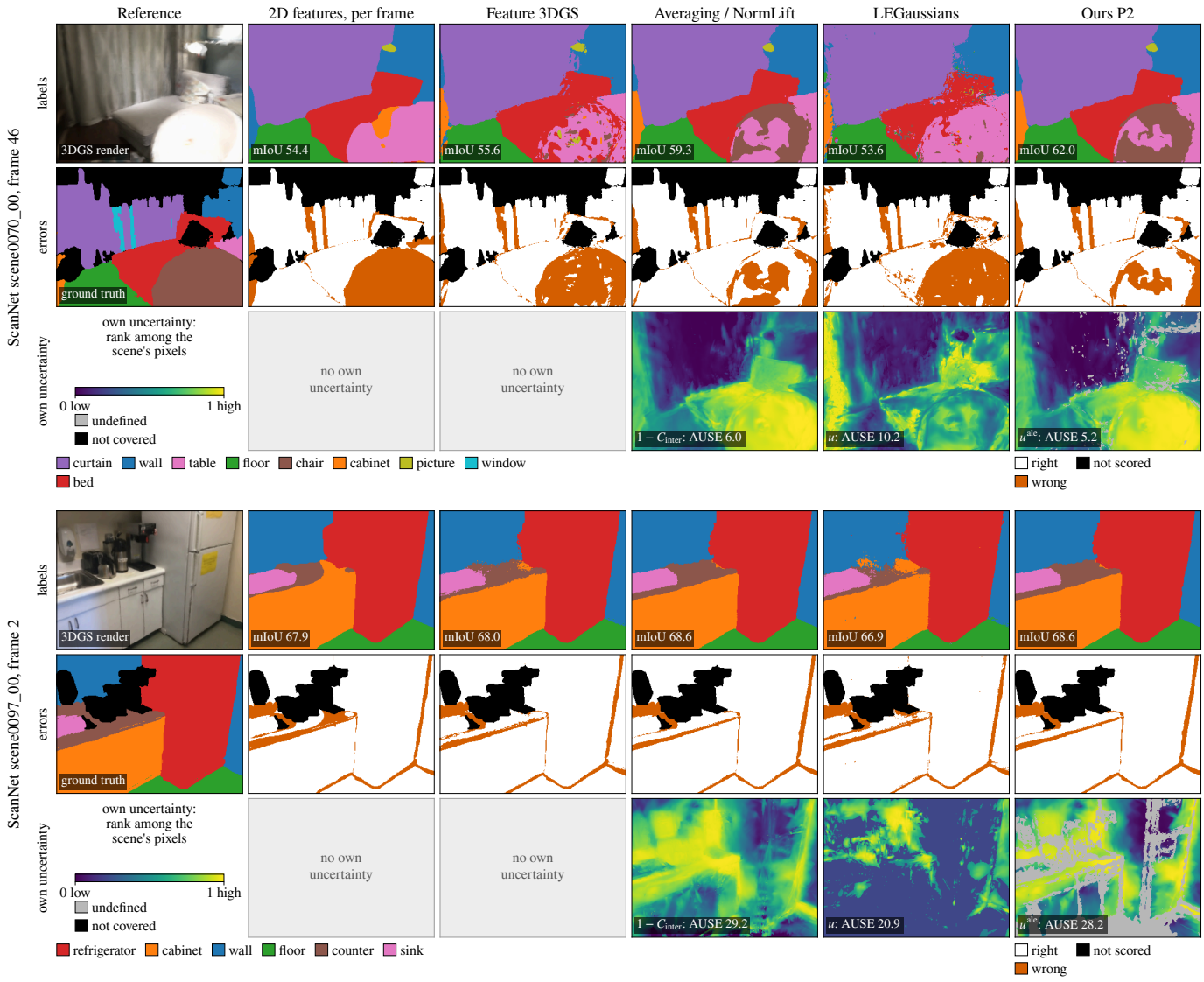


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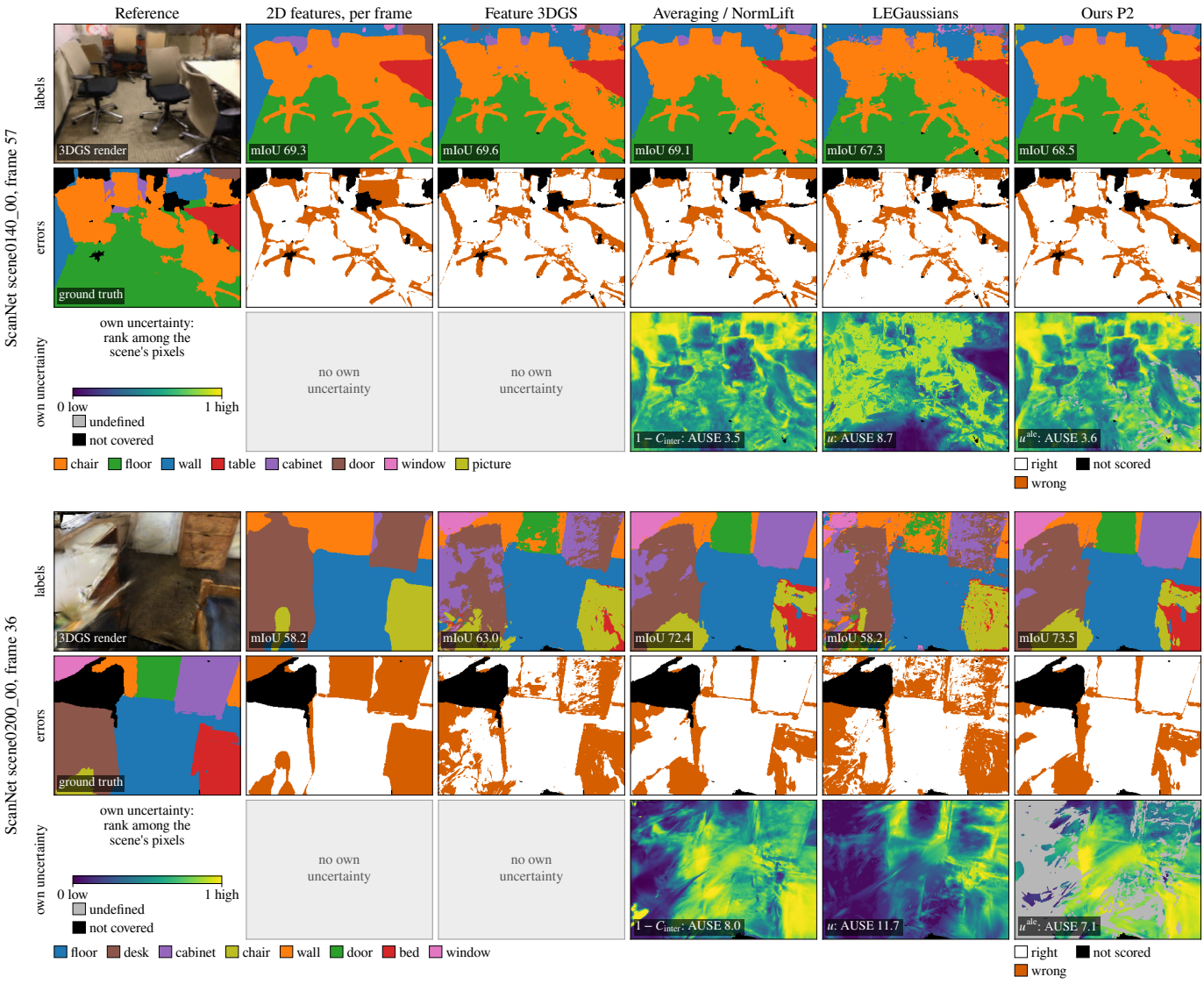


Fig. S10. Continued from Fig. S6.

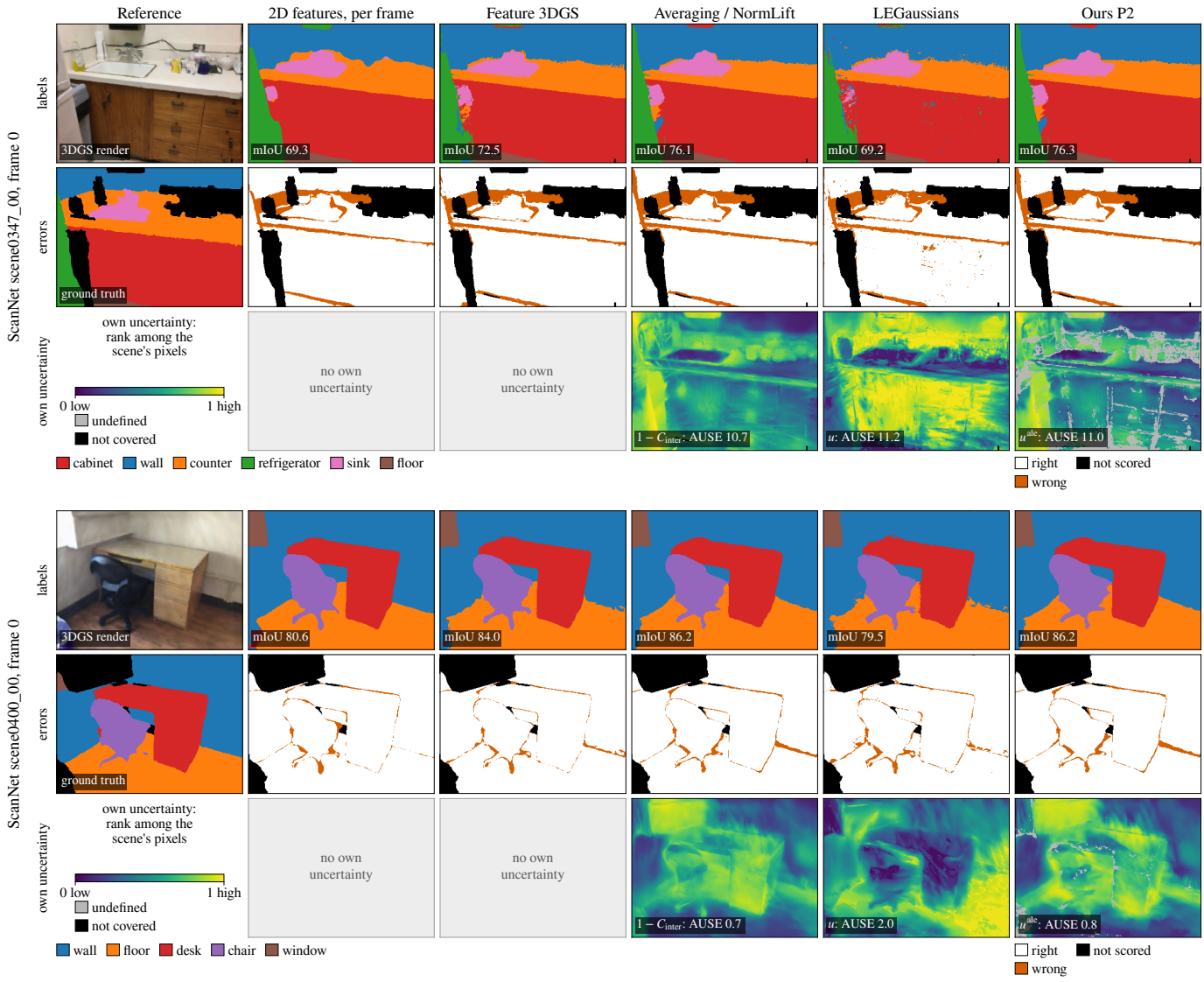


Fig. S11. Continued from Fig. S6.

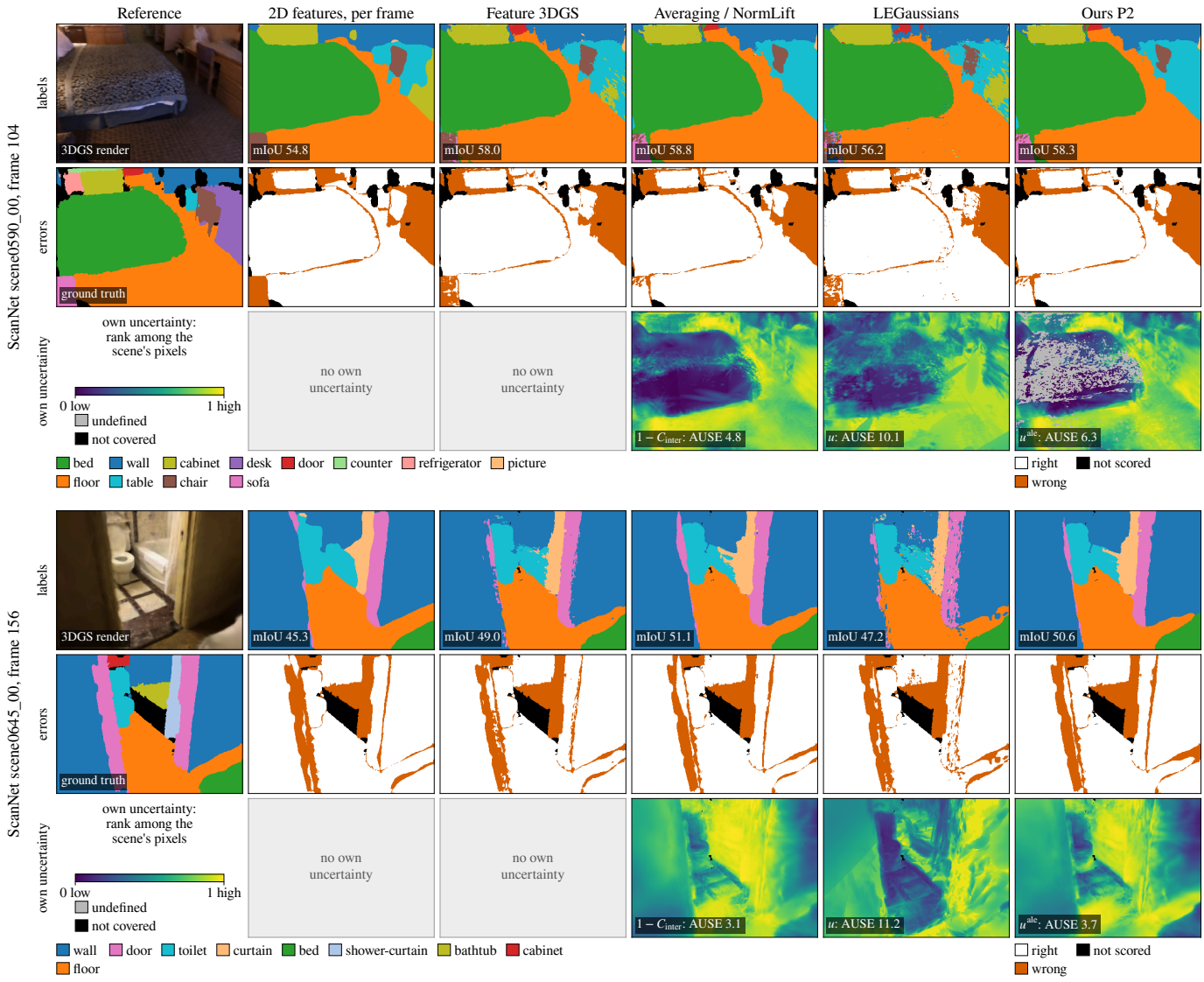


Fig. S12. Continued from Fig. S6.

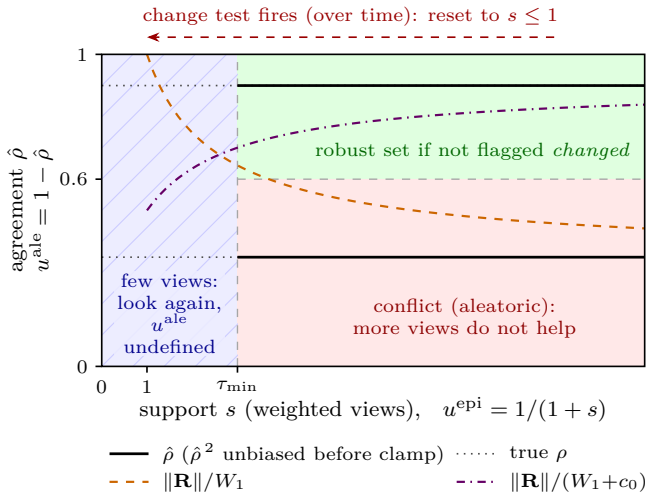


Fig. S13. Support s (weighted views) against agreement $\hat{\rho}$ of one Gaussian. The curves are root-mean-square values for two Gaussians with true agreement ρ (dotted). The mean length $\|\mathbf{R}\|/W_1$ makes the conflicting one look consistent at few views, and the shrunk $\|\mathbf{R}\|/(W_1+c_0)$ makes the consistent one look conflicting. Our $\hat{\rho}$ ($\hat{\rho}^2$ is unbiased before the clamp) stays close to ρ. A fired change test resets s to at most one view.

TABLE S6
P2 COEFFICIENTS FITTED BY NLL ON THE VALIDATION SCENES, AND THE TEMPERATURE β OF P0.

Dataset	β_0	γ_e	γ_a	γ_u	β (P0)
Replica	45.6	+0.36	+11.83	+0.055	36.0
ScanNet	34.6	-0.80	-2.06	+0.142	39.2

TABLE S7

BUILD TIME PER SCANNET SCENE IN SECONDS, MEDIAN (MIN–MAX) OVER THE TEN SCENES. TOP: PER STEP. BOTTOM: END TO END WITH THE SHARED 3D-GS FIT AND LSEG, AND THE PER-SCENE RATIO TO OURS.

Method	Step	Seconds
Ours, averaging, NormLift	field update, one pass (view hooks)	5.6 (3.0–27.4)
Every method	LSeg extraction	13.5 (5.7–48.4)
Every method	3DGS map fit	324.3 (266.6–694.5)
Feature 3DGS (re-impl.)	feature training, fixed map	1267.5 (740.0–2037.0)
LEGaussians (re-impl.)	feature training, fixed map	730.0 (385.0–1375.0)
Method	**Seconds**	**Ratio**
Ours, averaging, NormLift	346 (282–716)	reference
Feature 3DGS (re-impl.)	1614 (1051–2748)	4.0 (3.0–6.6)
LEGaussians (re-impl.)	1078 (782–2086)	2.9 (1.8–5.3)

TABLE S8

REGISTERED LATER ATTEMPTS. VALIDATION SCENES SELECT; THE TEST SCENES RUN ONCE AFTER THE SELECTION. STALE ANSWERS: SHARE OF THE REMOVED OBJECT'S COVERED PIXELS STILL ANSWERING ITS QUERY (%).

Attempt	Validation	Test
Vote predictive from the same posterior (with a P1 prior grid and P2 on P1)	Replica pooled NLL: P2 0.800, best alternative 0.809; P2 kept on both datasets	not selected; disclosed second look on Replica: vote predictive ECE 6.3, NLL 1.06, AURC 16.4 (P2: 8.7, 1.00, 12.2)
Pooling each Gaussian's statistics over its 3D neighbors (24 Gaussian kernels)	Replica: no kernel beats P2; ScanNet: a kernel selected (NLL 0.403 vs 0.410)	ScanNet NLL 0.535 vs 0.509 (P2) and 0.512 (averaging); ECE 6.6 vs 6.1, 6.2; AURC 5.4 vs 5.0, 4.4
Fused detectors fire the reset (our test with the feature and depth residuals)	stale answers 24.1 vs 24.8 (ours); Gaussian AP of the firing score 85.7 vs 60.3	stale answers 26.0 vs 20.7, lower in 3 of 10 scenes; firing AP 79.3 vs 53.4; FA 3.7 vs 6.2
Fusing the map with the current frame, weighted by u^{epi} and u^{ale}	Replica: every fused row +1.4 mIoU; ours best NLL	fused rows change mIoU by at most 0.4; ours worse than a constant weight in 11 of 14 scenes (mIoU, NLL)
Stronger extractor for every row (RADSeg, ProxyCLIP ViT-H, Trident)	2D mIoU Replica / ScanNet: LSeg 30.8 / 67.5, RADSeg 34.0 / 43.1, ProxyCLIP 26.1 / 38.0, Trident 29.4 / 35.9	not run: no candidate gained on both datasets

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